# Lab 2 - Advanced Machine Learning

## Clustering

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# General information

- **Assignment** : Alone or in pairs, you will code the algorithms you learnt in Ścikit-learn formalism’
- **Due** : Each assignment has to be sent at most 7 days after the lab session at pierre.houdouin@centralesupelec.fr
- **Grading** : There are 5 lab sessions, each lab session is worth 4 points, the average will count for half of your final grade
- **Questions** : If you have questions, comments or feedbacks about the lab session, feel free to contact me by email

# Lesson recap

| | type | n_clusters | Objective | Algorithm | Robust to | Clusters |
|---|---|---|---|---|---|---|
| K-Means | partitional | hardcoded | $\min_{\delta_{ik}, c_k} \delta_{ik} \sum_{k=1}^{K} \sum_{i=1}^{m} \left\| x^i - c_k \right\|^2$<br>cluster sets (location and assign.) within-cluster variance | alternatively assign points to clusters, recompute clusters as center-of-points | | Points that are near.. |
| Agglomerative Single-Linkage | hierarchical (bottom-up: merge) | given by…<br>'cutoff' ε | - | sequentially compute distance (e.g. min) between clusters and merge the two nearest clusters, until you end up with one cluster. | init | …nearest |
| DBSCAN | partitional | given by…<br>'cutoff' ε<br>density minPts | - | Identify core points as having at least minPts in their ε-neighborhood.<br>Their connected components on the ε-neighbor graph make the clusters.<br>Non-core points either join an ε-nearby cluster, else are noise. | …and outliers, noise | … and in dense regions |
| HDBSCAN | hierarchical (top-down: split) | given by…<br>~~'cutoff' ε~~<br>density minPts | - | 1. Build complete graph weighted by specific metric that penalizes sparsity*<br>2. Extract the minimum spanning tree<br>3. Construct a cluster hierarchy of connected components by removing heaviest edges<br>4. Condense the cluster hierarchy based on a min. cluster size before merge (less is noise)<br>5. Extract the clusters with long antecedance (robust to cutoff) in the condensed tree : tunes ε for each cluster. | …and n_clusters | … that are not easily split |

*for two 'close' points,
clamp their distance to that to the farthest Minpts neighbor.

# Assignment plan

- **Part 1** : Use scikit-learn K-means
- **Part 2** : Implement your own Agglomerative Single-Linkage algorithm
- **Part 3** : Use scikit-learn DBSCAN
- **Part 4** : Use scikit-learn HDBSCAN
- **Part 5** : Applications to a Mars image